Research on Forward Collision Warning System Based on Connected Vehicle V2V Communication

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***Abstract*—In this paper, a Forward Collision Warning (FCW) system based on Dedicated Shortrange Communications (DSRC) technology is built to improve vehicle driving safety and reduce traffic accidents. When the Time to Collision (TTC) and the Collision Avoidance Range (CAR) system calculated show that the vehicle is in danger of collision, the driver will be alerted via the voice and visual interface. In order to evaluate the safety and reliability of the warning system, several groups of field experiments were carried out to test the system performance, including the positioning accuracy of Differential GPS (DGPS), the communication quality of DSRC and the trigger of FCW system. We confirm the operation of the FCW system.**

Keywords—Intelligent Transportation System, Forward Collision Warning, Connected Vehicle, Vehicle to Vehicle Communication

I. INTRODUCTION

With the rapid increase of car ownership in China, the road safety situation has become increasingly severe. According to the data of the World Health Organization (WHO) survey report, more than 260,000 people die from traffic accidents in China every year. In all traffic accidents, the proportion of accidents caused by forward collision is more than 60 percent [1]. In these forward collision accidents, the vast majority of them are due to the driver's failure to detect the dangerous vehicles in front of him in time and his inadequate response. Therefore, if the driver can be warned effectively before the traffic accident happens so that he can get a more adequate response time to take corresponding measures, the accident can be effectively avoided.

Intelligent network interconnection of vehicles is an important way to improve road safety. It aims to enhance vehicle's environmental awareness through various intelligent sensors, and realize Vehicle to Everything communication (V2X, including V2V, V2I, V2P and V2N) and data interaction through DSRC, LTE-V and other connected vehicle technologies to help vehicles identify potential dangers and react in a timely manner.

Based on the connected vehicle environment, Berkeley's Sengupta et al. studied a coordinated collision system based on the relative position and speed of the car and the neighboring car [2]. Qin Wenhu et al. designed a robust rear-end warning system. In view of the dangers of false alarms and false negatives, the neural network is used to train the data, which has a good effect [3]. Gumaste et al. used V2V communication technology to predict the location of the vehicle's impending collision, forming a collision area of the vehicle, helping the driver to adjust the vehicle's motion state in time to avoid collision [4]. Although these methods have been proved by simulation to achieve good performance when the positioning error is small enough, the accuracy of the system's ranging accuracy still has room for improvement.

The FCW system in connected vehicle environment usually uses lidar and high-definition camera as sensors to identify and locate the front vehicles. Although these sensors have high accuracy, they can not work robustly in bad weather such as rain, snow and fog, and can not effectively identify dangerous vehicles in visual blind areas [5]. On the other hand, civilian GPS can identify dangerous vehicles in blind areas, but because it has only 10m-level positioning accuracy, it is not enough to support the reliability requirements of safety applications such as FCW and other applications closely related to safety. However, the accuracy of DGPS can reach decimeter level or even centimeter level. Using the longitude and latitude data of DGPS to calculate the relative distance between vehicles makes it possible to apply this technology to the FCW system.

Based on the connected vehicle V2V environment, combined with DSRC communication technology and high-precision positioning technology of DGPS, a FCW system is developed to promote the practical application of forward collision warning. DSRC is a technology specially used for vehicle communication. It can provide high-speed data transmission, and can guarantee low delay of communication

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link and ensure system reliability. Therefore, this forward collision warning system selects DSRC communication technology to realize Vehicle-to-Vehicle (V2V) communication [6]. By installing On-Board Equipment (OBE) on the vehicle, the system transmits the FCW data of the Remote Vehicle (RV, refers to the background vehicle that can broadcast V2X regularly in conjunction with HV) to the Host Vehicle (HV, refers to the vehicle equipped with OBE and running FCW application program) through V2V communication, and calculates the real-time TTC and CAR between vehicles based on positioning data and vehicle status data to judge the collision risk. If the system determines the forward collision risk, the driver will be warned by visual and voice alarm through the visual interface of mobile APP.

In addition, to verify the reliability of the FCW system, this paper tests the scenarios specified in T/CSAE 53 standard (Cooperative intelligent transportation system; vehicular communication; application layer specification and data exchange standard), such as the front vehicle is still, the front vehicle is moving slowly, the front vehicle is slowing down and the front vehicle is moving on the opposite lane. At the same time, the DGPS positioning accuracy and V2V communication quality (delay and packet loss rate) which affect the system security are tested.

II. The Connected Vehicle Platform in Wuhan University of Technology

The FCW system is based on the connected vehicle test platform of Wuhan University of Technology. As shown in Fig. 1, our test platform is mainly composed of OBE, Roadside Equipment (RSE), Traffic Signal Controller (TSC) and mobile devices. OBE can obtain its own vehicle status data in real time and broadcasting continuously through DSRC technology. The RSE installed on the road side is connected to the TSC via TCP/IP protocol. RSE can acquire the Signal Phase and Timing (SPaT) message from the TSC, and broadcast it to the OBE in the communication range. As a display terminal, the mobile device is connected to the OBE through a WI-FI network, and displays valuable information acquired by the OBE on the screen. In this paper, OBE, which is mainly used, is an intelligent device developed by China Huali Smart Ways Technology Company. The two-way real-time communication between OBEs can be realized by DSRC technology. The connection of notebook computer and OBE through network line can realize real-time data transmission and storage.

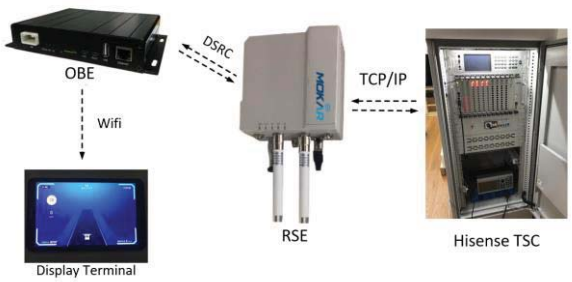

Fig. 1. Architecture of connected vehicle platform

III. Functional Details of the FCW System

FCW means that when the HV is driving in the lane and there is a risk of rear-end collision with the RV on the same lane in front of it, the application of FCW will warn the HV driver, assist the driver to avoid or mitigate forward collision and improve road safety.

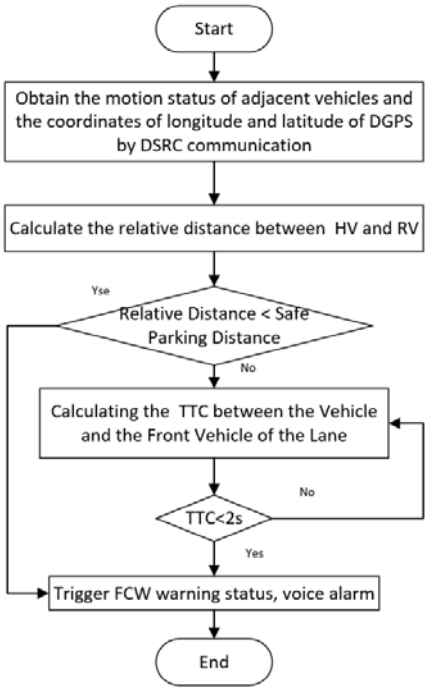

Fig. 2. FCW system flow chart

The basic working principle and algorithm flow of FCW are relatively simple compared with other security applications. This is to reduce the computational load of the system, improve the efficiency and robustness of the system. The key steps and processes of system implementation are shown in Fig. 2. Firstly, the OBE of HV analyses the received RV messages, filters the RV which is stationary, slow or decelerated driving in front of the HV in the same lane, and then calculates the TTC and anti-collision distance between the HV and each RV within a certain distance, and screens out the threatened vehicles with potential collision risk with HV. If there are multiple threatened vehicles, the most urgent threatened vehicle will be selected, and finally the system carries out the corresponding collision warning to the driver through HMI.

A. *Vehicles Broadcast Message*

In our FCW system, the transmission of messages between vehicles is mainly realized by broadcasting Basic Safety Message (BSM) packets outward through OBE. BSM is currently the most widely used application layer message, which is used to exchange security status data between vehicles. Vehicles broadcast and receive BSM messages to inform the surrounding vehicles of their real-time status, but also to obtain the status of the surrounding vehicles. FCW system can identify OBE-equipped vehicles within a certain range in front of HV driving direction according to the position, speed and acceleration information of BSM package obtained by analysis. Of course, as a security auxiliary application, reliability and security are the most important, so the communication quality of BSM packet

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transmission needs to be strictly controlled. In the following chapters, we design field tests to test the packet loss rate and unidirectional transmission delay of BSM messages transmitted by OBE devices [7].

B. Exclude the Vehicles in Opposite Lanes

Although the V2V communication of DSRC can realize the communication between vehicles equipped with OBE, the vehicles in front of the vehicle can also be screened out. However, for vehicles coming from opposite lanes, because they are also in front of HV. If there is no distinction between vehicles coming from opposite lanes, there will be false alarm and reduce the reliability of the system. Therefore, in the two-way two-lane scenario, we judge whether two target vehicles are located in the same lane according to the vectors of the direction of the vehicle [8].

As shown in Fig. 3, according to the current and previous position of HV, we can get the direction vector of HV. Similarly, according to the position data transmitted from RV by V2V communication, the RV driving direction vector can also be obtained. If the angle theta obtained by vector translation is in the range of [0,90], it is considered that the two target vehicles are moving in the same direction. If the angle theta is beyond this range, RV is considered to be in the opposite lane. At this time, the possibility of two vehicles without collision risk can be judged without further calculation of the warning strategy. The system will ignore such incoming vehicles without triggering the warning.

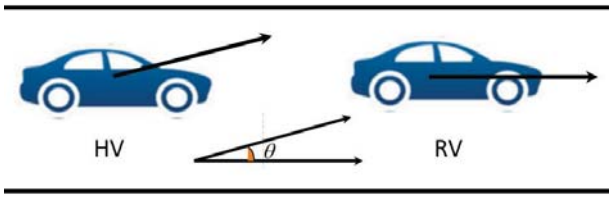

Fig. 3 Schematic diagram of algorithm for excluding the vehicles in opposite lanes

Through this filtering method, the FCW system can avoid a considerable part of the calculation of non-collision dangerous scenarios, thus greatly saving the computing resources of early warning strategy and improving the operational efficiency of FCW system.

C. Distance Calculation

After excluding the opposite vehicle, the relative distance between each potential dangerous vehicle and HV needs to be calculated. FCW system calculates this distance according to the longitude and latitude coordinates of DGPS of HV and RV. The calculation formula is as follows:

$$D_r = \pi \cdot R \cdot \frac{1}{180 \cdot \sin \omega A \cdot \sin \omega B + \cos \omega A \cdot \cos \omega B \cdot \cos(jB - jA)} \quad (1)$$

In the formula, jA and ωA are the longitude and latitude of point A, and jB and ωB are the longitude and latitude of point B, respectively. It is assumed here that the earth is a perfect sphere with radius R, the position of HV is A (jA, ωA), and the position of RV is B (jB, ωB), then the theoretical relative distance of the two vehicles is the arc length of AB.

As a security assistant application, FCW system has a high requirement for the relative distance between HV and RV to ensure the security of application, so it is necessary to test the positioning accuracy of DGPS. This part of the field test experiment will be described in detail in the following chapters.

D. Trigger Algorithms

The trigger condition of FCW is usually the TTC or the distance of collision avoidance [9]. These two indicators can effectively identify vehicles with high collision risk. In this paper, two indicators are all used as trigger conditions to improve the security of early warning. As long as one condition is met, early warning will be triggered. The trigger condition formula of FCW is as follows:

$$TTC \le 2s, \ D_r \le d_w \quad (2)$$

$$TTC = \frac{D_r}{v_{HV} - v_{RV}} \quad (3)$$

$$d_w = v_{HV} * t_d + (v_{HV} - v_{RV}) * t_s + (\frac{v_{HV}^2 - v_{RV}^2}{2 * a_{\max}}) + \mathrm{d}_0 \quad (4)$$

In the formula, v_{HV} is the speed of HV, v_{RV} is the speed of RV, d_w is the warning distance, t_d is the driver delay, t_s is the delay of FCW system, $a_{\max}$ is maximum acceleration when the vehicle is braking , d_0 is the minimum safe distance.

When the TTC calculated by the system is less than 2 seconds or the relative distance between the two vehicles is less than the warning distance, the FCW will be triggered. TTC is set to 2 seconds because it is a generally accepted value to trigger such a system [10]. When TTC is less than 1 second, collision is inevitable. At the same time, the FCW system is still used in the vehicle experiment. In order to ensure the safety of the test vehicle in the experimental process, the TTC time of 2 seconds can also meet the safety needs.

E. Alert Driver

Fig. 4 Screenshots of FCW system triggered

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When the FCW warning is triggered, the OBE will send an alert to the mobile device connected with WIFI. The alert is visually displayed on the screen and accompanied by a voice announcement. In addition to the forward collision warning messages, the driver can also see the current speed, driving time and other vehicle status information on the terminal. The terminal display screenshot after triggering the early warning is shown in Fig. 4.

IV. FIELD TEST SETUP

In order to verify the safety and reliability of our FCW system, we carried out field tests using the built connected vehicle experimental platform for several problems mentioned above. Firstly, whether the positioning accuracy of DGPS can meet the needs of FCW. Secondly, whether the communication quality of DSRC V2V communication between OBEs (considering packet loss rate and delay) is good enough. We all know that in the case of high-speed vehicles, excessive delay will cause huge security risks, which is unacceptable. Finally, whether the complete FCW system can be triggered normally and whether it will produce false alarm to the vehicles in opposite lanes.

The site of the field test is located on a road adjacent to intramural athletic facilities in Yujiatou Campus of Wuhan University of Technology, as shown in Fig. 5. It is a north-south two-lane road with a total length of 670 meters. The reason for choosing this road instead of urban roads is to minimize other interference factors while ensuring the safety of road tests and ensuring repeatability and comparability of results.

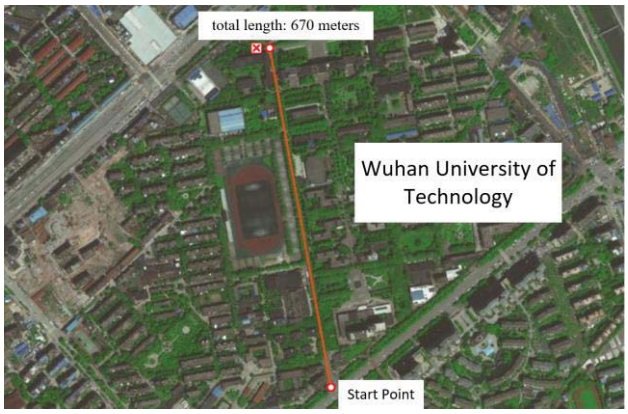

Fig. 5 Testing sites for field tests

A. DGPS Positioning Accuracy Test

In order to test the positioning accuracy of DGPS, we used two OBEs for static test. As shown in Fig. 6, OBE1 is deployed at point A, and we test the relative distance between OBE1 and OBE2 calculated by the FCW system every ten meters [11]. Under each working condition, the FCW application report log data acquired in 1 minute period is counted and each working condition is subjected to 5 repeated tests, and the average value of the test results is taken as the value of the relative distance.The maximum test distance of 200 meters is chosen because the distance of 200 meters is enough for forward collision warning. The relative distance is calculated automatically by the latitude and longitude coordinates of the two points acquired by the system. We use a laptop to connect with OBE2 by Wi-Fi network to remote login to the device. We obtain the relative distance by saving the corresponding report log of the FCW application. The positioning accuracy of DGPS is evaluated by comparing the relative distance with the actual distance.

The dynamic DGPS accuracy test experiment here is not implemented due to the lack of means to obtain real-time real distance between vehicles.

Fig. 6 Schematic diagram of point selection for field test

B. Static DSRC Communication Quality Test

In order to test the DSRC communication delay of FCW system, we also used two OBE devices. The location of the equipment is the same as that of the DGPS precision test. Similarly, within a distance of 0 to 200 meters, every 10 meters interval is tested to obtain data by saving the corresponding log file of OBE2.

In the test, the packet sending rate was set to send one BSM packet every 100 milliseconds, and 600 packets obtained in one minute were counted. Each working condition was tested repeatedly for 5 times, and the average value of the test results was taken as the final result of the communication delay and the packet loss rate.

a) Communication Delay Test

For communication delay testing, we need to get the time when OBE1 sends a BSM packet to OBE2. The general idea is to send a BSM packet through OBE1, record the time stamp of sending the packet, and then record the time stamp of receiving the BSM packet on the OBE2 device. The difference between the two times is the delay of one communication transmission. However, the problem is that there is an error in the acquisition time of OBE from GPS (although the error is in milliseconds), and the communication delay of DSRC is also in milliseconds. After testing, it may even occur that the sending time is later than the receiving time, which obviously can not accurately test the DSRC communication delay of OBE equipment.

In order to measure the DSRC communication delay, we turn to test the loop time of communication, that is, the total time for OBE1 send BSM packets to OBE2 and OBE2 send another BSM packet to OBE1 immediately after OBE2 receives the packet. Half of the loopback time is the communication delay we need.Of course, the time of logical reaction inside OBE2 is neglected here. In order to achieve this goal, we rewrite the Software Development Kit (SDK) code of OBE device, and change the BSM packet sending logic of OBE2. OBE2 send a BSM packet only after receiving a BSM packet instead of broadcasting the BSM packet continuously. Fig. 7 is a sample of the data we

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obtained. The data of time has been marked in the picture by a red rectangle.

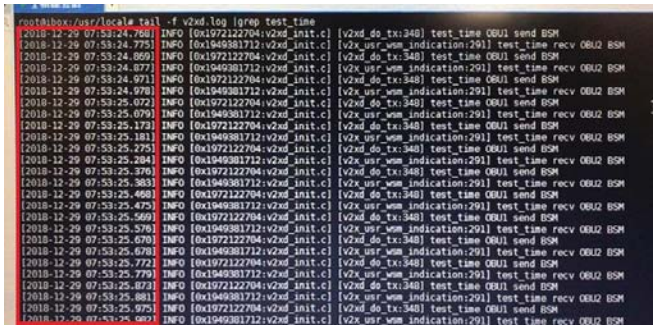

Fig. 7 Samples of loopback time data obtained

b) Packet Loss Rate Test

For the test of packet loss rate, we calculate the packet loss rate of communication by counting the number of BSM packets sent by OBE1 and the number of BSM packets received by 0BE2 in one minute under various conditions. These data are also obtained by saving the OBE log file.

C. *Dynamic DSRC Communication Quality Test*

Through static DSRC communication quality test, we can get the basic performance of OBE equipment. However, in general, vehicles that open the FCW system are generally in motion, so test of dynamic DSRC communication quality is also essential.

For the dynamic DSRC communication quality test, we used two vehicles equipped with OBE. One test vehicle is the Toyota Corolla, which has a displacement of 1.2 L and a length of 4.6 meters. The other test vehicle is the Dongfeng Fengshen AX7 commercial vehicle which has a displacement of 1.6 L and a length of 4.7 meters. In the test, Toyota Corolla was used as the RV, Dongfeng Fengshen AX7 was used as the HV, and the HV was equipped with a tablet computer equipped with the FCW application.

The method of obtaining communication delay and packet loss rate data is the same as that of static experiment. The difference is that in the dynamic experiment, we use the HV speed as a variable and test the DSRC communication quality at 10-50 Km/h with an interval of 10 Km/h. Each working condition was tested repeatedly for 5 times, and the average value of the test results was taken as the final result of the communication delay and the packet loss rate.

D. *FCW System Trigger Test*

We tested several scenarios of RV, such as keeping still, driving slowly and slowing down. In addition, in order to verify whether the algorithm for excluding the vehicles in the opposite lane is effective, we also conducted a false alarm test on the scene where the RV was driving in the opposite lane. The purpose of this experiment is to verify the accuracy of the FCW algorithm in the connected vehicle V2V environment and evaluate the performance of the warning system.

a) Scene Test for RV's Keeping Still

The RV remains stationary 200 meters ahead of the HV, ie the speed of the RV is 0 Km/h. The HV starts the FCW system and drives to the RV at five speeds of 10 Km/h, 20 Km/h, 30 Km/h, 40 Km/h, and 50 Km/h. The scene data is obtained by saving the OBE log file of the HV.

b) Scene Test for RV's Driving Slowly

The RV keeps driving at a low speed and at a constant speed 200 meters ahead of the HV. The HV starts the FCW system and drives to the RV at a constant speed higher than the RV. Their speeds are v_{HV} =30 Km/h v_{RV} =10 Km/h, v_{HV} =40 Km/h v_{RV} =10 Km/h, v_{HV} =40 Km/h, v_{RV} =20 Km/h respectively. The data is obtained by saving the OBE log file of the HV.

c) Scene Test for RV's Slowing down

The RV maintains a uniform motion of 20Km/h at 200 meters ahead of the HV. The HV starts the FCW system and drives to the RV at a constant speed of 40Km/h. The RV suddenly decelerates at a distance of about 30m from the HV. The scene data is obtained by saving the OBE log file of the HV.

d) False Alarm Test

The false alarm test mainly tests the FCWS's ability to screen the target object and whether it can filter non-threatening objects (such as vehicles traveling in the opposite lane). If the false alarm rate of the FCW system is too high, frequent warnings to the driver will not only cause the driver to be disgusted with the alarm, but also cause the driver to burn out the alarm sound. Even if there is a real danger, the driver can't make a timely response.

For the false alarm test, the RV maintains a constant speed at different speeds in the opposite lane while The HV is moving forward. We need to observe whether the FCW system generates false alarms.

V. Result and Discussion

In order to evaluate the reliability and safety of the FCW system, we conducted field experiments for DGPS positioning accuracy, static DRSC communication quality, dynamic DSRC communication quality, and triggering of the FCW system. The system performance was evaluated by collecting data such as vehicle speed, relative distance, delay, and packet loss rate during the experiment.

A. *Influence of Distance on DGPS Positioning Accuracy*

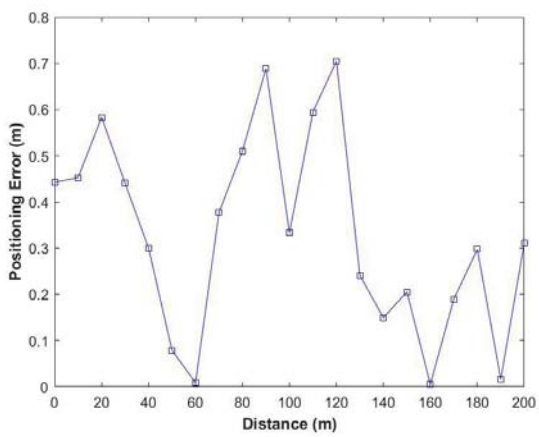

Fig. 8 The relationship diagram of positioning error and distance

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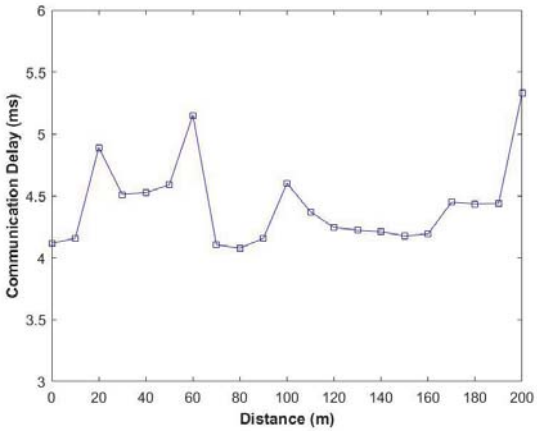
The relationship between positioning error and distance is shown in Fig. 8. The numerical value of the positioning error is the absolute value of average relative distance of each point minus the real distance [12]. From the figure, we can see that in the range of 0-200 meters, with the increase of the distance, the positioning accuracy of DGPS has no obvious change trend, the average error is 0.33 meters, and the maximum error is about 0.7 meters. This shows that the accuracy of DGPS has been greatly improved compared to the positioning accuracy of 10 meters for ordinary civilian GPS. However, the maximum error of 0.7 meters is still somewhat inadequate for the FCW system, which emphasizes safety. Some studies have found that for static positioning, the accuracy of DGPS can reach 2-5cm [13]. The reason why our DGPS can't reach such high precision in the field test is that the accuracy of our DGPS receiver itself is sub-meter level and cannot reach the centimeter level. Secondly, there are many trees on both sides of the test section, which also has some influence on the accuracy of DGPS.

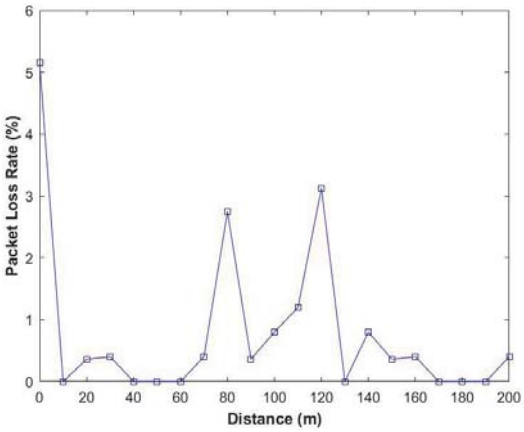
Of course, we only tested the accuracy of static DGPS because of the lack of means to capture the true distance between vehicles during vehicle motion. The 0.7 meters error is measured at the standstill of the device and does not fully reflect the positioning error of the vehicle under motion. In the follow-up study we plan to further test the dynamic positioning error.

Under the current test conditions, due to the existence of DGPS error, we increase the minimum safe distance parameter in the alarm distance by 0.7 meters to ensure the safety of the FCW system.

B. Influence of V2V Communication Distance on Communication Quality



(a) The relationship diagram of communication delay and distance



(b) The relationship diagram of packet loss rate and distance

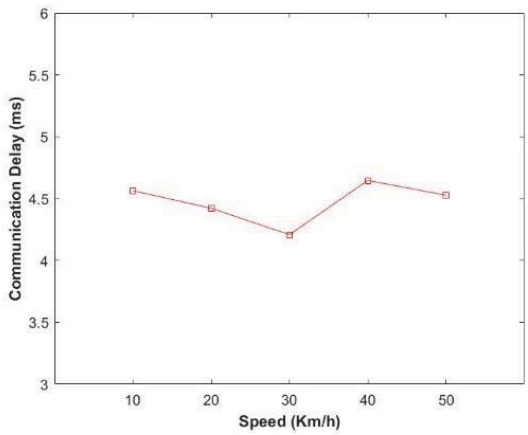
Fig. 9 Static Communication quality performances

The relationship between the communication delay and the V2V communication distance is shown in Fig. 9(a). In the range of 0-200 meters, the measured maximum delay is 5.3 ms. The increase in distance will hardly cause a significant increase in delay, and the delay is basically stable between 4.1-4.8 ms, and in a few cases it will slightly exceed 5 ms.

The relationship between the packet loss rate and the V2V communication distance is shown in Fig. 9(b). In the communication range of 0-200 meters, the packet loss rate of the DSRC transmission BSM packet has some fluctuations, but basically remains below 3%. Among them, when the distance is 0 meters, the packet loss rate is abnormally high, reaching 5%. This is due to the close distance between the two OBEs and the change of coupling mode [14]. Generally, there is no overlap between two vehicles, so this situation can be ignored.

Taken together, within a range of 200 meters, communication delays below 5ms and a maximum packet loss rate of 3% are a good attribute for FCW systems. The millisecond communication delay makes the FCW system not need to consume more time, which provides guarantee for system security. The maximum packet loss rate is 3%, especially between the critical distances of 0-40 meters triggered by the FCW system. The packet loss rate less than 0.4% indicates the reliability of the network.

C. Influence of Vehiclle Speed on Communication Quality



(a) The relationship diagram of communication delay and speed

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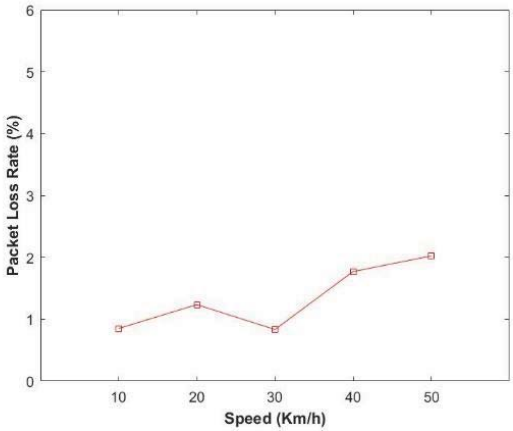

(b) The relationship diagram of packet loss rate and speed

Fig. 10 Dynamic Communication quality performances

The relationship between communication delay and vehicle speed is shown in Fig. 10(a). In the speed range of 10-50 Km/h, the measured communication delay fluctuates slightly between 4.2 and 4.6ms. The increase in speed has no obvious effect on communication delay. In addition, the fluctuation range of 4.2-4.6 ms is included in the range of 4.1-4.8 ms in the distance test, which indicates that the two factors of distance and speed have no obvious influence on the V2V communication delay.

The relationship between the packet loss rate and the vehicle's motion speed is shown in Figure 10(b). In the speed range of 10-50 Km/h, the measured packet loss rate is between 0.8% and 2%. As the speed increases, the packet loss rate increases. Combined with the packet loss rate data in the distance test, it shows that the distance to the packet loss rate is not obvious in the range of 0-200 meters, and the vehicle running speed is positively correlated with the packet loss rate.

D. Triggering of FCW System

After three sets of system trigger test experiments, the system trigger log generated by the OBU device is redirected, and the data such as the vehicle speed, the relative distance during the alarm, and the warning threshold (ie, the safe parking distance) in each group of experiments are obtained, as shown in Tab 1.

Tab. 1 FCW system scene data

Scene preset speed（Km/h）		Actual speed（Km/h）		Warning threshold (m)	Relative distance during alarm (m)
		Speed of the HV	Speed of the RV		
RV remains stationary	10-0	10.131955	0.013855	8.129802	8.017379
	20-0	22.148832	0.016787	12.960639	12.591772
	30-0	29.683269	0.023502	16.936577	16.359859
	40-0	38.859633	0.017256	22.765378	22.441691
	50-0	52.42985	0.010297	33.369877	32.641827
RV drives slowly	30-10	32.567279	9.503137	17.807735	17.679601
	40-10	41.086188	11.458483	23.180836	22.522443
	40-20	41.909807	20.216566	21.753468	21.632950
RV slows down	40-20	26.691556	8.611763	14.554955	14.404290

It can be seen from Tab 1 that in the three scenes of the front vehicle keeps still, the front vehicle drives slowly and the front vehicle slows down, the corresponding scene data is captured. This shows that the FCW system can trigger normally under the corresponding conditions of these scenarios. In addition, we can also see that the system's early warning threshold is dynamically changed according to the vehicle speed of the front and rear vehicles. The HV will only trigger an alarm when the relative distance between the vehicles is less than the warning threshold.

As for the false alarm field test, FCW alarms were not triggered in several groups of vehicle tests, so the relevant alarm data was not captured. But this is precisely the exclusion of the opposite lane vehicle algorithm is effective.

VI. CONCLUSION

In this paper, we build a FCW system based on the experimental platform of Wuhan University of Technology. In order to evaluate the safety and reliability of the system, we designed the corresponding field experiments to obtain the relevant data. After analyzing the corresponding data and results, we draw the following conclusions: The first conclusion is that the static positioning accuracy of DGPS can basically meet the requirements of the FCW system. The second conclusion is that DSRC V2X communication is a low-latency, high-reliability communication method with good performance on the FCW system, which increases the security and reliability of the system. The third conclusion is that the FCW system we built can realize the correct alarms of multiple scenes such as the front vehicle stays still, the front vehicle drives slowly and the front vehicle slows down, and the algorithm for excluding the Vehicles in Opposite Lanes can work normally, and can avoid false alarms.

The realization of FCW in the connected vehicle environment can effectively reduce the occurrence of traffic accidents, especially rear-end accidents, but we still have a lot of work to do. For example, the positioning accuracy of the vehicle under motion needs to be further tested. The next step is to install a radar on the test vehicles to obtain real-time distance data between the vehicles to help test the dynamic DGPS positioning accuracy. In addition, DGPS positioning accuracy can also be improved by replacing DGPS receivers with better performance. All in all, the security and reliability of the system application will determine the application level of the connected vehicles. How to improve the security and reliability of the FCW system will be our continuous consideration in the future.

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